

Passive cardiac shear-wave elastography: processing methods explored, as space-time plots

shearWaveProcessing – `study/analysis/passive_methods_atlas.py`

24 September 2026

What this document is

The passive (valve-closure) shear-wave work compared many processing choices: which motion quantity to show, the temporal band-pass, spatial and temporal smoothing, M-line averaging, directional filtering, clutter and motion handling, and recipes from the literature. The decisions were recorded in the repo's docs (`docs/passive_search.md`, `HANDOFF.md`), but mostly on one acquisition and often through automatic scores. This document shows the choices *side by side on the same in-vivo windows*, as space-time (M-mode) plots, so they can be judged by eye. **No automatically fitted speed is shown anywhere**; it is not what we trust (`docs/passive_speed_estimation.md`).

All data are in vivo (buffer 4, the ~ 926 Hz diverging-wave stream of the S5-1 sequence). The phantom has no passive shear waves.

How to read the panels

- **Axes.** Horizontal: time relative to the detected event peak, in ms (dotted line at 0). Vertical: distance along the analysed part of the M-line, in mm. So a wave travelling along the septum is a *tilted* band, and motion of the whole wall at once is a *vertical* band.
- **How much tilt to expect.** A wave at 3 ms^{-1} crosses a 20 mm line in about 7 ms. That is small on a 130 ms axis, so Figure 2 repeats the key comparison zoomed to ± 25 ms.
- **Colour.** Each panel has its own symmetric colour scale (99th percentile of $|\cdot|$), so the plots show pattern, not amplitude.
- **Columns.** Seven windows, always in the same order. Six were scored *clear* in the hand-labelled set: three aortic valve closures (AVC) and three mitral valve closures (MVC). The last column was scored *none*: it shows what processing makes of a window with no visible wave. **A method that makes the last column look like a wave is manufacturing structure.**
- **Timing.** The detected event peak and the visible wave do not always coincide: in C000000003 the wave is ~ 40 ms before the peak.

Baseline recipe

Every family starts from view A of `configs/passive.yaml` and changes one thing. View A:

- Loupas lag-1 estimator, frame-to-frame, cumulative displacement;
- 10–150 Hz zero-phase Butterworth band-pass;

- Gaussian spatial smoothing, $\sigma = 0.6 \text{ mm axial} \times 1.2 \text{ mm lateral}$;
- 3-frame moving mean;
- 5 parallel M-lines 0.5 mm apart, averaged;
- no directional filter.

The exceptions are the literature recipes and the three production views, which are complete recipes. The IQ is cropped to the M-line box plus 8 mm before processing.

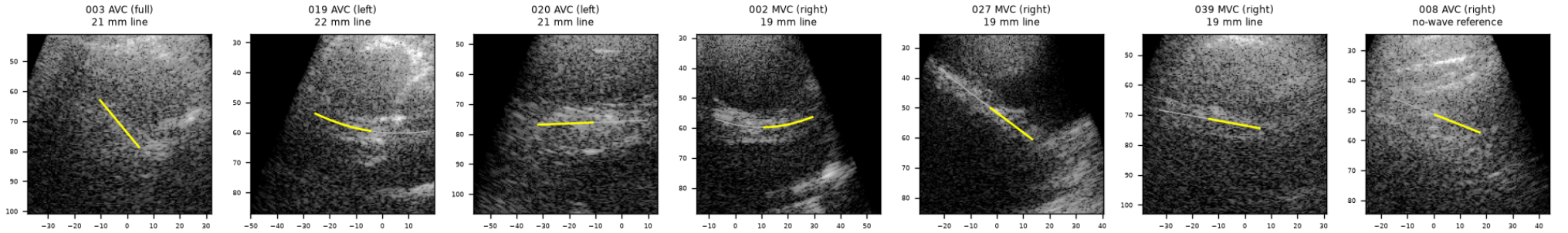


Figure 1: The seven windows: the buffer-1 B-mode frame at the event's cardiac phase, on which the per-event M-line was drawn. Yellow: the part of the line analysed (the part hand-labelled); grey: the rest of the line. M-lines are 19–22 mm.

1 Motion quantity: displacement, velocity, acceleration

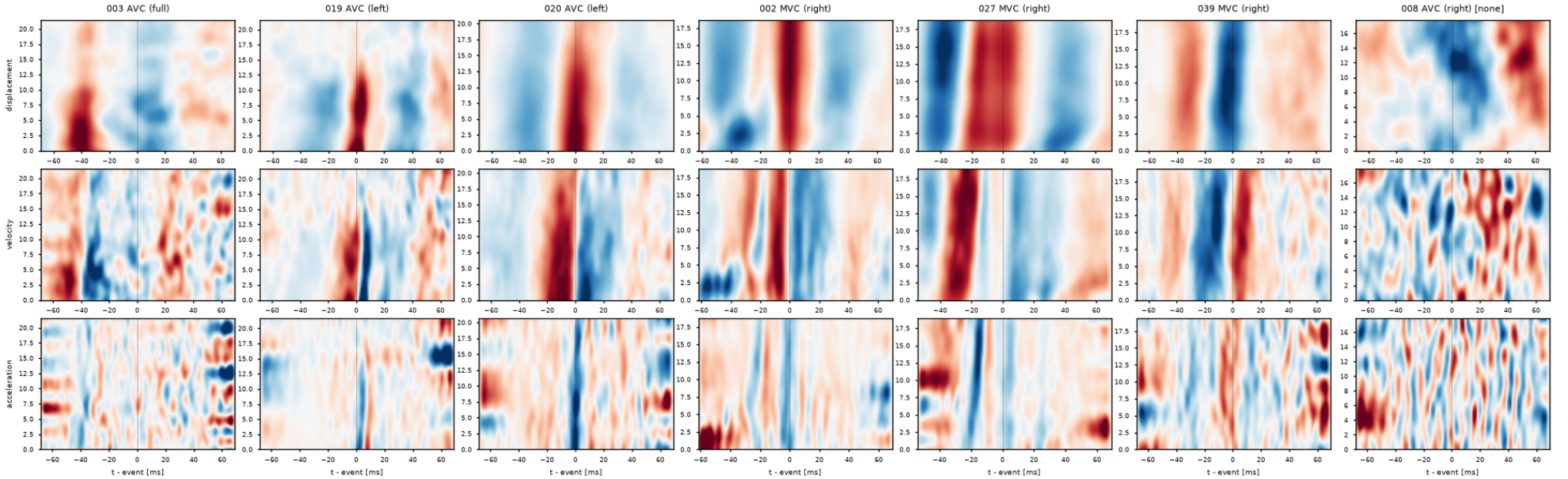
What varies. The same frame-to-frame estimate shown as cumulative displacement, as particle velocity, or as acceleration (the temporal derivatives).

Literature. Velocity is used by Keijzer et al. (2019, 2020). Acceleration is used by Petrescu et al., Santos et al. (2019), Strachinaru et al. and Espeland et al. (2024). Nobody reports cumulative displacement. The wave is dispersive, so the three weight different frequencies and give different apparent speeds.

Earlier conclusion. Displacement was chosen for visualisation (cleanest), velocity was noisier, and acceleration had the sharpest front but was noisiest.

What the atlas shows.

- **Displacement:** smooth but broad bands, roughly 15 ms to 25 ms wide. At this width a tilt of a few ms is hard to read.
- **Velocity:** a bipolar front about half as wide. The tilt becomes visible in C000000019, 020 and 002 (Figure 2).
- **Acceleration:** the narrowest front exactly at the event (019, 020, 027 – in 027 the clearest tilted line of all). But outside the event it is noise-dominated, and in 003 and 039 the front is lost.
- **The no-wave window** is incoherent in velocity and acceleration. Displacement turns it into smooth broad blobs that are easier to misread.



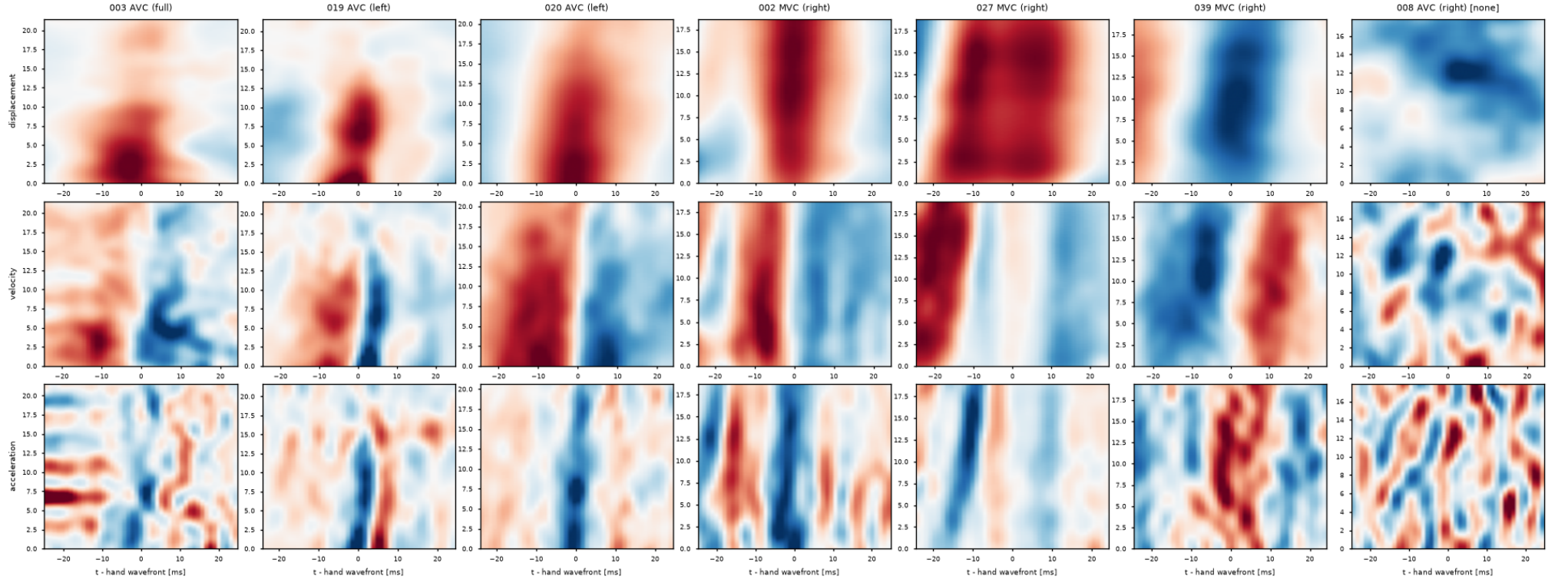


Figure 2: The quantity comparison of the previous page zoomed to ± 25 ms around the midpoint of the hand-drawn wavefront (event peak for the no-wave window). Tilted fronts are readable in velocity (019, 020, 002, 003) and in acceleration (027). Displacement bands are too wide to show the slope well.

2 Temporal band-pass

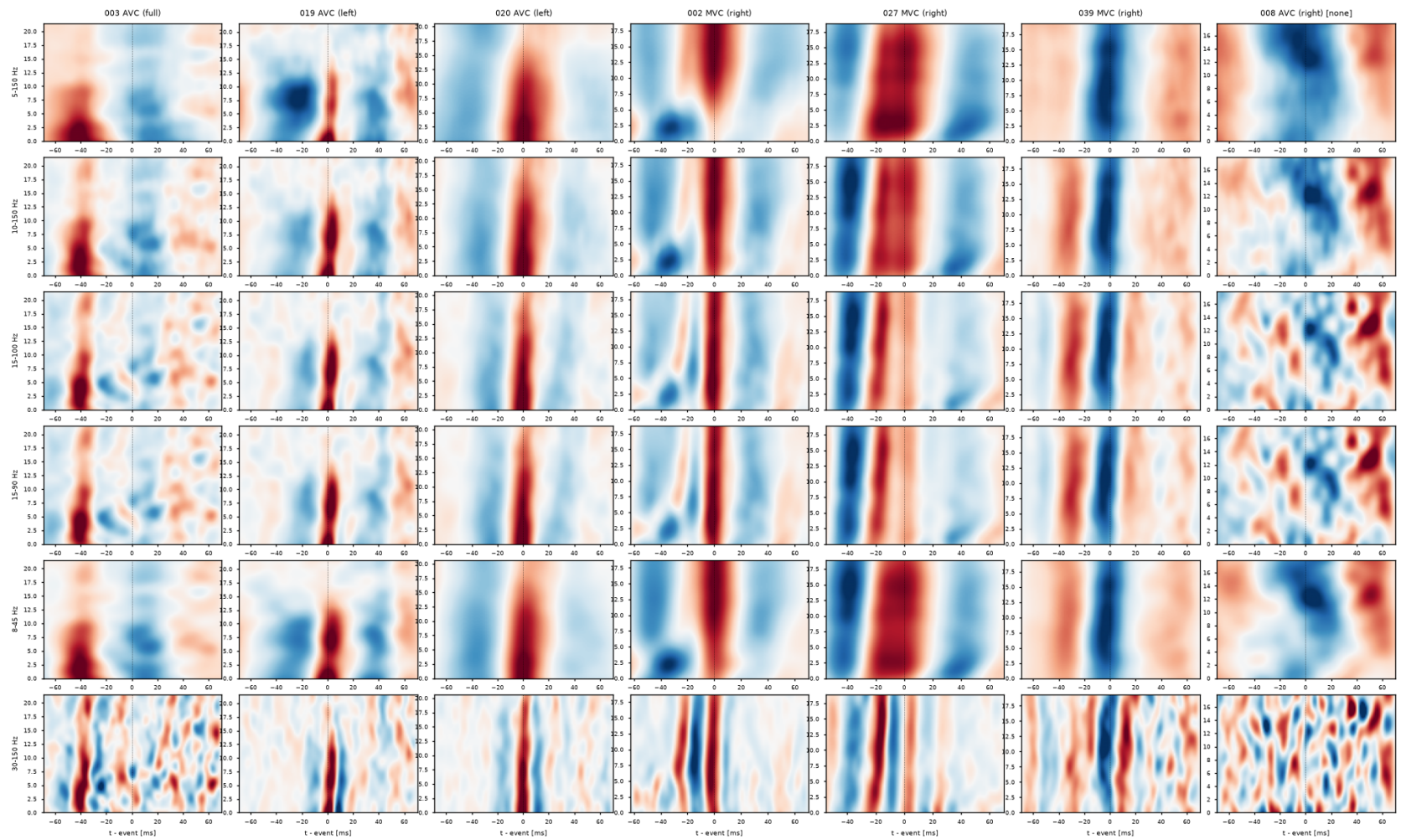
What varies. The zero-phase Butterworth band-pass that separates the wave from bulk wall motion (low corner) and from noise (high corner).

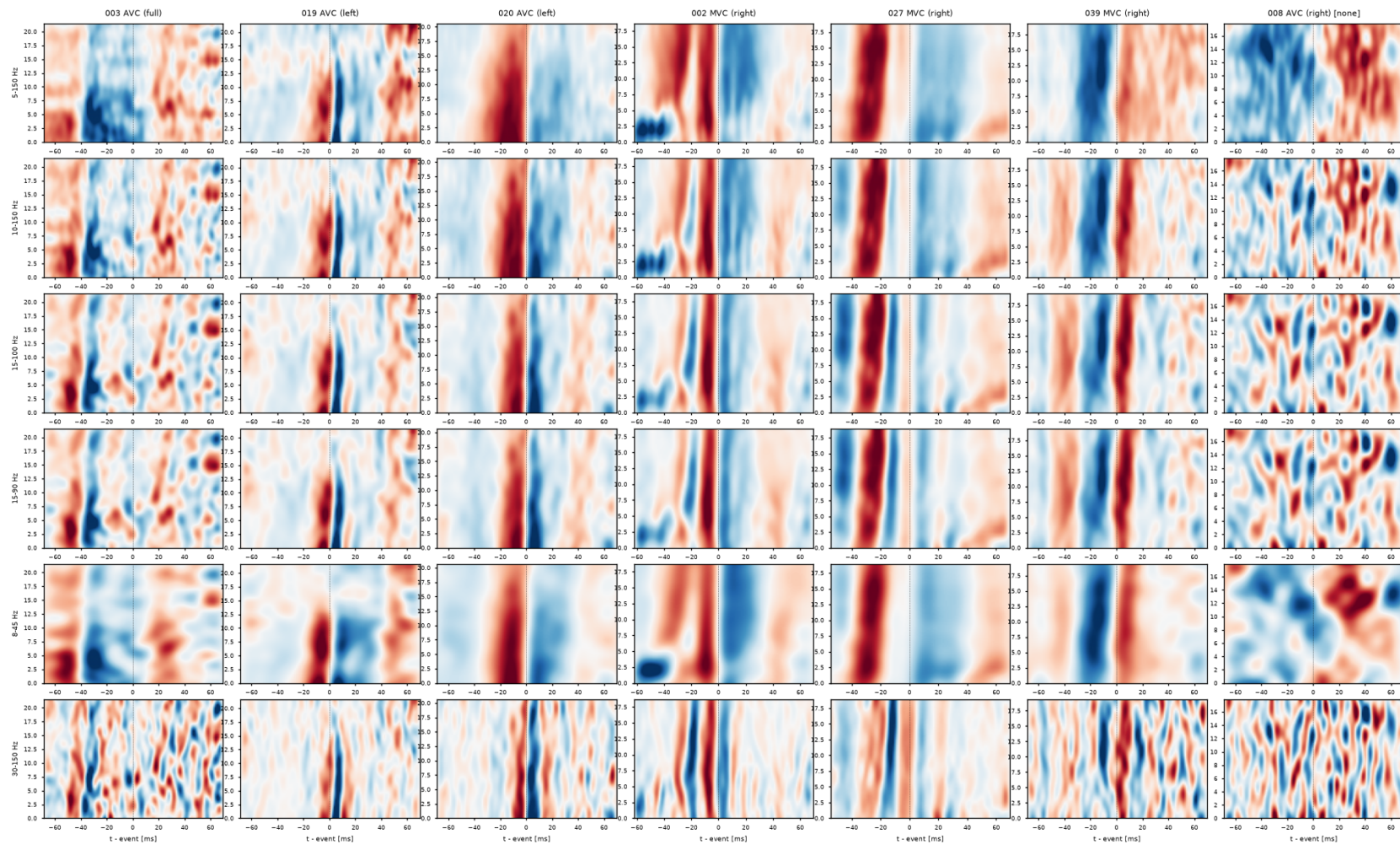
- 15–100 Hz is Keijzer’s band and the Caenen review’s band for MVC waves.
- 15–90 Hz is view C.
- 8–45 Hz is the narrowing tried on 2026-09-22 and reverted.
- 30–150 Hz tests a higher low corner.

Earlier conclusion. The low corner must be at least 10–20 Hz, because 5 Hz lets the flat bulk band through. The band content ends near 90 Hz (f99), and 8–45 Hz cut real signal.

What the atlas shows (displacement first, then velocity).

- **5–150 Hz** leaves large slow blobs (e.g. 019 at -20 ms, 020 and 027 broad): bulk wall motion.
- **10–150 Hz** removes most of it.
- **15–100 and 15–90 Hz** give the narrowest clean fronts in both quantities. In velocity they are the cleanest panels of the whole atlas, and the no-wave window stays incoherent.
- **8–45 Hz** smears every front into a broad band.
- **30–150 Hz** turns each event into several narrow parallel bands (ringing): the low corner now sits inside the wave’s own spectrum, and the no-wave window becomes fine texture.





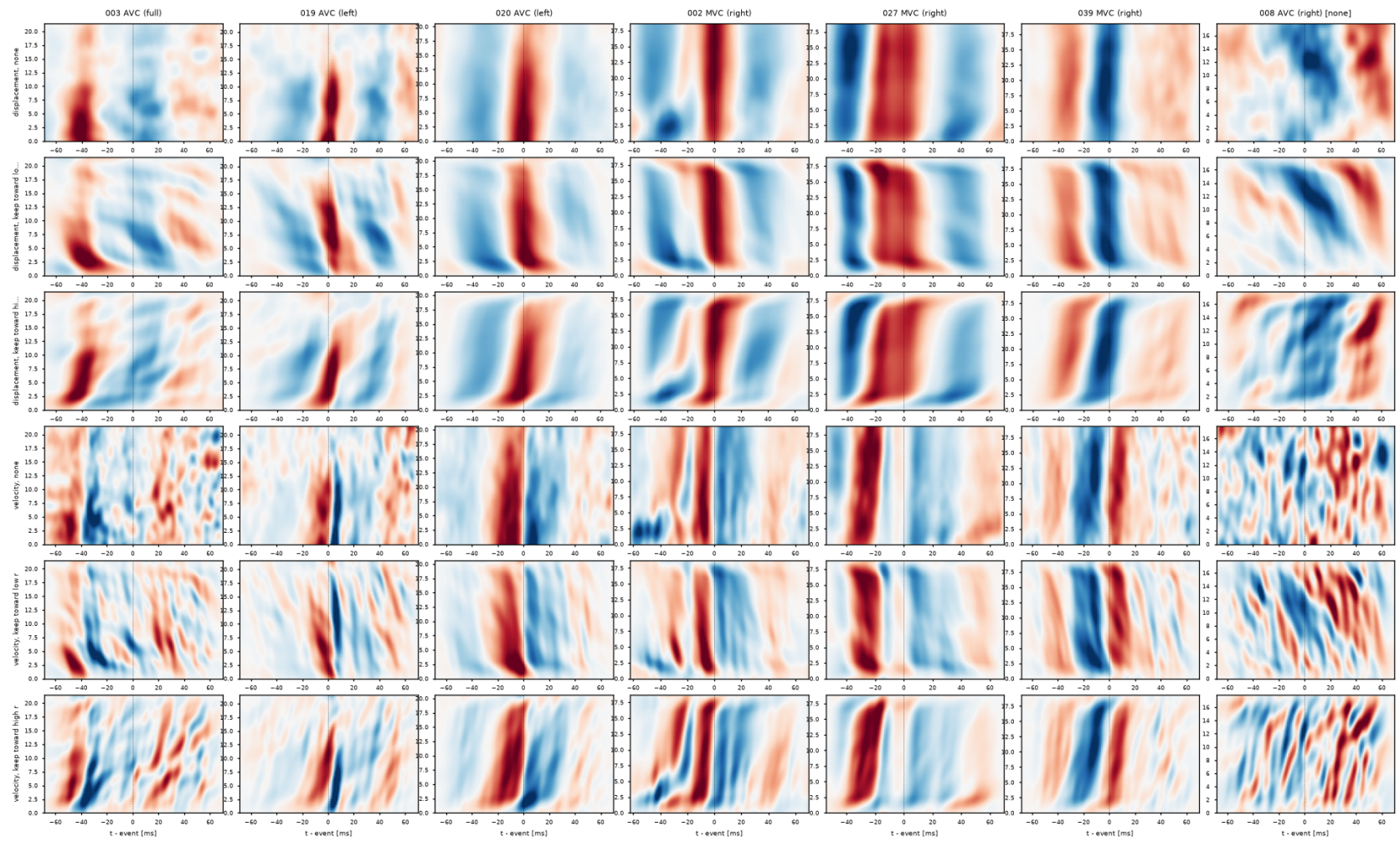
3 Directional (k - ω) filter

What varies. The filter keeps only waves travelling one way along the line (toward low or toward high r). It is standard for ARF shear waves (Song et al. 2016). For natural waves none of the papers use one.

Earlier conclusion. Keep it off for passive: it added banding and biased the slope. The synthetic tests of 2026-09-24 also show it biases the active speed high.

What the atlas shows. It imposes a tilt on everything.

- In the no-wave window it produces clean, regular tilted stripes – a convincing-looking wave that is not there.
- In the real windows the apparent tilt follows the direction chosen: the same event tilts one way with “toward low r ” and the other way with “toward high r ” (019, 020, 039).
- The filter therefore decides the direction instead of measuring it. It must stay off whenever the direction is not known in advance.



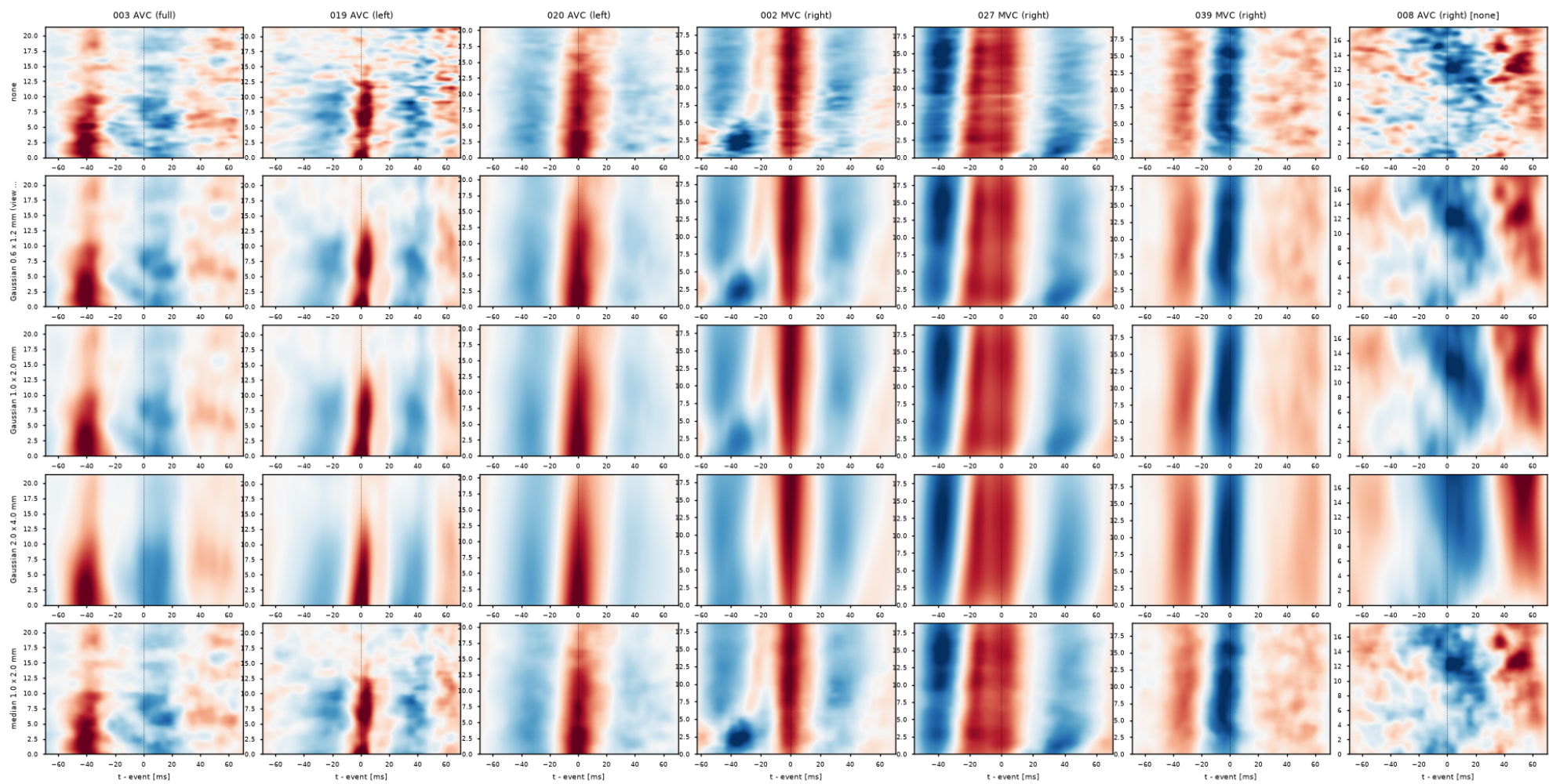
4 Spatial smoothing

What varies. Per-frame smoothing of the displacement field before sampling the M-line.

Earlier conclusion. Light Gaussian (0.6–1.0 mm) is best; median is similar or slightly worse.

What the atlas shows.

- **None:** fine horizontal stripes along r (speckle decorrelation between depths).
- **Gaussian 0.6×1.2 mm** removes them.
- **Gaussian 1.0×2.0 mm** changes little beyond that.
- **Gaussian 2×4 mm** makes the fronts look clean. But it turns the no-wave window into smooth coherent bands as well: too much.
- **Median 1×2 mm** sits between none and Gaussian and keeps some stripes.

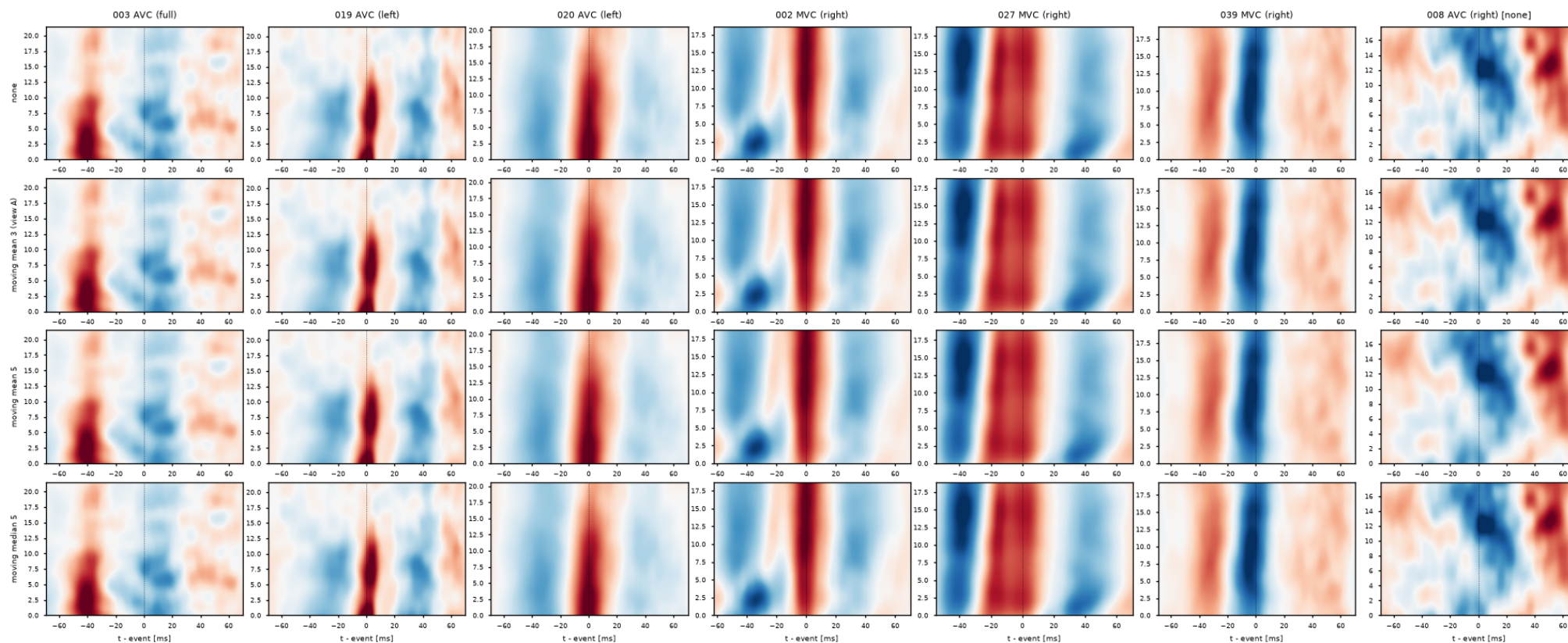


5 Temporal smoothing

What varies. A moving mean or median over 3–5 frames (3–5 ms).

Earlier conclusion. Second-order.

What the atlas shows. Almost no visible difference between none, mean 3, mean 5 and median 5. The 150 Hz low-pass corner already limits the bandwidth more than these windows do. Temporal smoothing matters for velocity and acceleration (derivatives amplify noise), not for band-passed displacement.



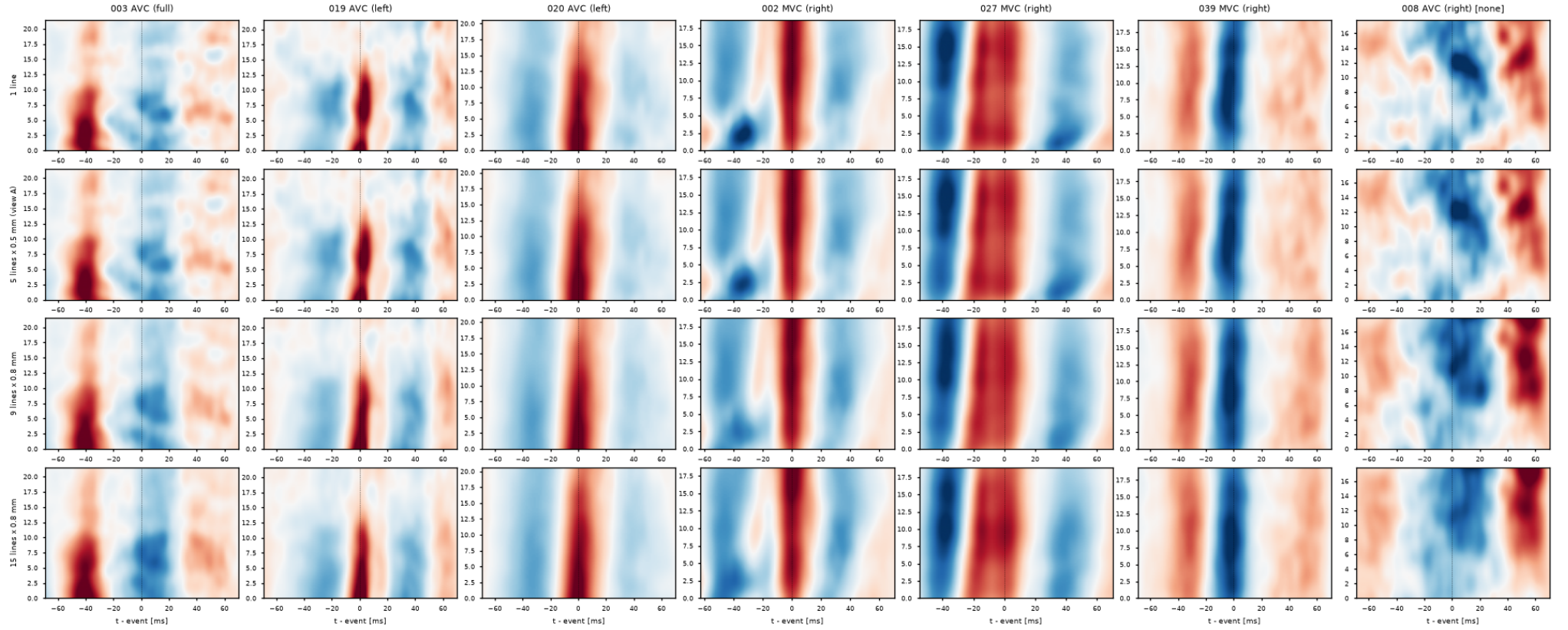
6 M-line averaging

What varies. How many parallel copies of the M-line, offset along its normal, are averaged.

Earlier conclusion. Few lines. One line is best for the sharp acceleration front; more lines help smoother quantities.

What the atlas shows.

- Little change from 1 to 9 lines in displacement.
- At 15 lines (a ± 5.6 mm band – more than half the wall thickness) the no-wave window becomes smoother and more band-like, and near the wall edges (003, 002) the pattern changes.



7 Motion and clutter handling

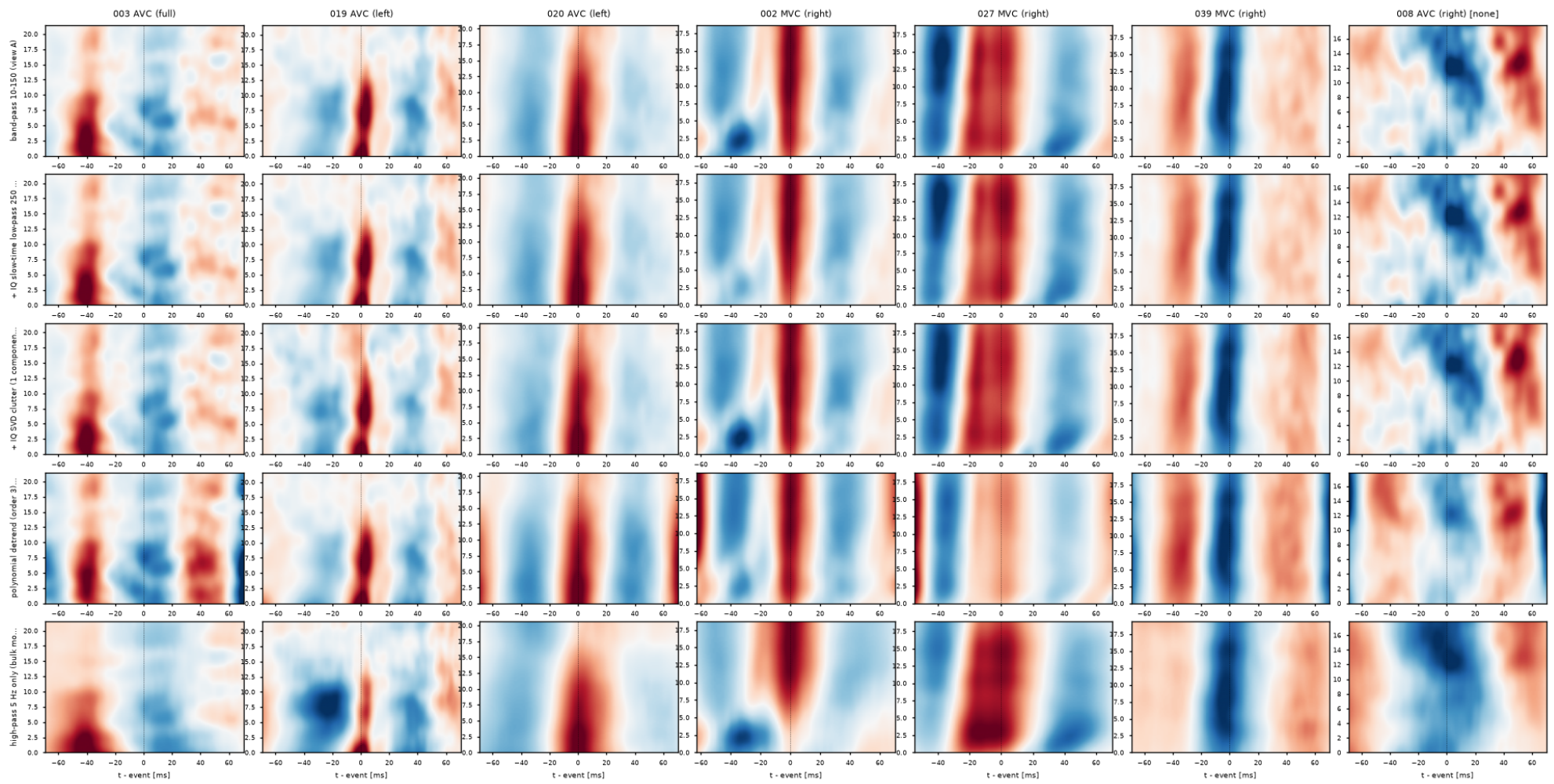
What varies. What removes the bulk wall motion:

- the band-pass alone (view A);
- plus an IQ slow-time low-pass at 250 Hz (Keijzer’s blood/noise filter);
- plus an SVD clutter filter on the IQ (one component, Demené-style);
- a polynomial detrend instead of the band-pass high side;
- or only a 5 Hz high-pass, which leaves the bulk motion in as a reference.

Earlier conclusion. The low corner of the band-pass is the decisive motion remover. SVD and polynomial detrending do not substitute for it, and the 250 Hz IQ low-pass washed out the wave in the search.

What the atlas shows.

- With the 10 Hz high-pass in place, the IQ low-pass and the SVD filter leave the panels essentially unchanged.
- The polynomial detrend adds bands at both window ends (the polynomial does not fit the edges) and keeps more low-frequency content.
- The 5 Hz-only row is dominated by broad bulk motion.

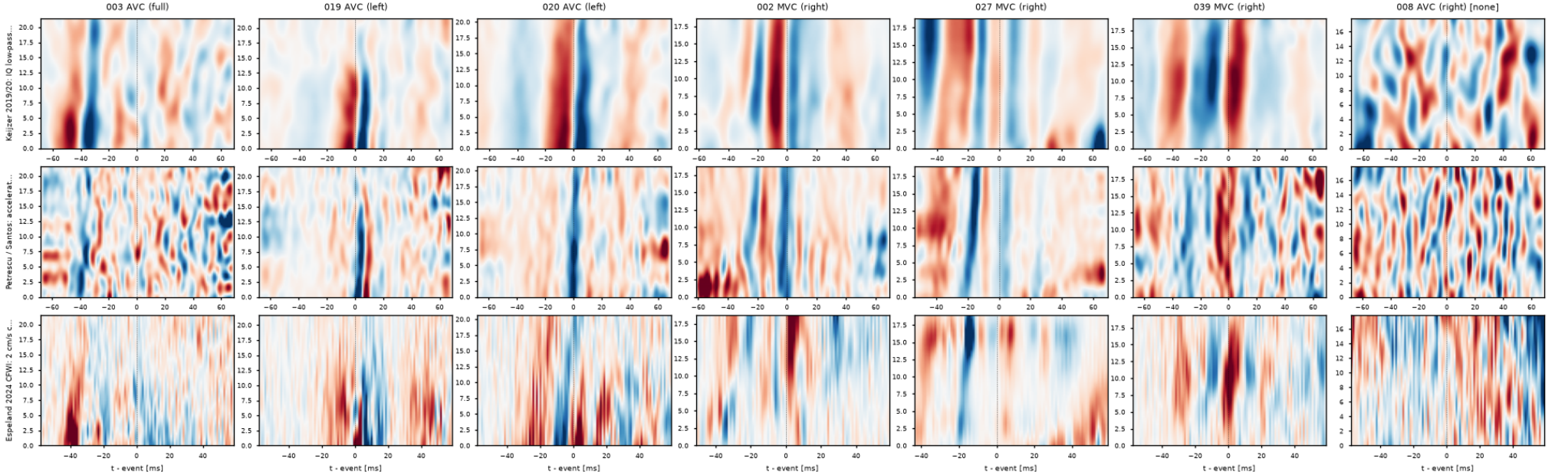


8 Recipes from the literature

- **Keijzer et al.** IQ slow-time low-pass at 250 Hz; lag-1 velocity with Gaussian smoothing of the autocorrelation before the angle (4 mm axial; their lateral 6.7 is approximated as 3 mm); 15–100 Hz band-pass (6th order).
- **Petrescu / Santos.** Acceleration, with a 3-frame moving mean on the velocity before differentiation.
- **Espeland et al. 2024 (clutter filter wave imaging, CFWI, PLAX view).** IQ high-pass at a 2 cm/s cutoff, envelope, temporal derivative. The first and last 10 frames are dropped (filter end transient).

What the atlas shows.

- **Keijzer’s velocity recipe** looks like the 15–100 Hz velocity rows: clean bipolar fronts, and an incoherent no-wave window.
- **The acceleration recipe** has sharp fronts at 019, 020 and 027 and is noise-dominated elsewhere.
- **CFWI** shows the event as a front in 002, 027 and 039, faintly in 019 and 020, with lower contrast and vertical streaks. Its automatic direction agreed with the hand picks only at chance level (`study/analysis/cfwi_benchmark.py`).



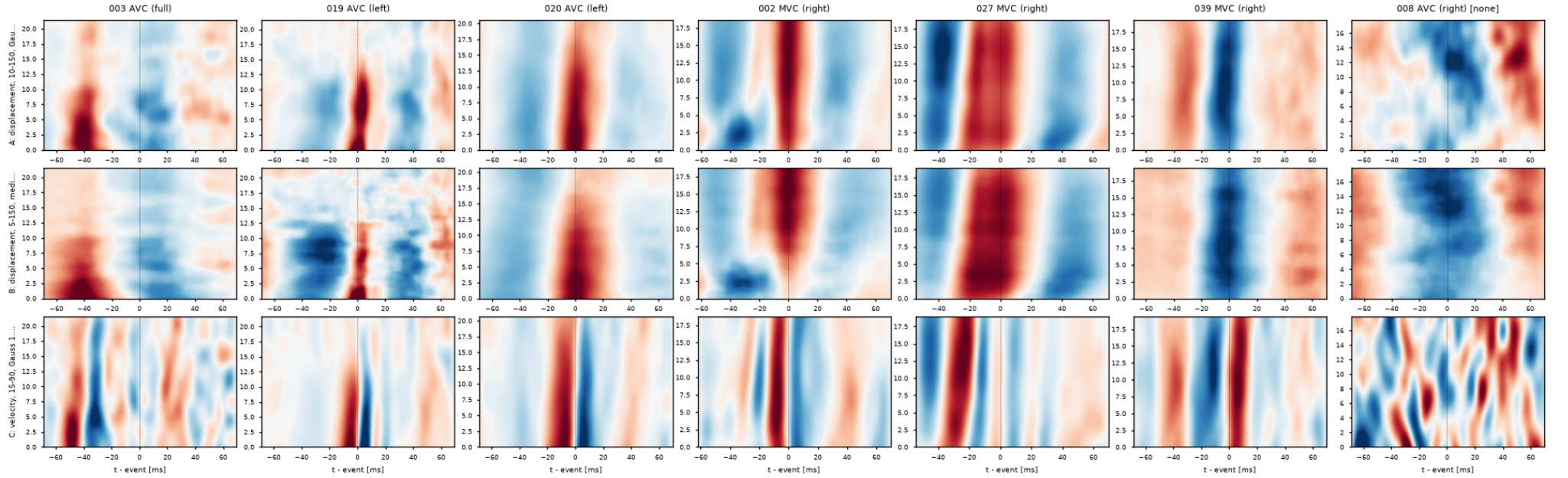
9 The three production views

`configs/passive.yaml` renders three deliberately different recipes per window:

- **A**: displacement, 10–150 Hz;
- **B**: displacement, 5–150 Hz, median 1×2 mm, no temporal smoothing, 9 lines;
- **C**: velocity, 15–90 Hz, Gaussian 1.0×2.0 mm, mean 5, 9 lines.

In these windows:

- **C** gives the sharpest, most readable fronts and the most clearly incoherent no-wave window.
- **B** carries bulk-motion blobs from its 5 Hz corner (019, 027) and median streaks.
- **A** is in between: smooth but broad.



10 Summary

choice	what the space-time plots show (7 windows)	suggestion for visual reading
quantity	displacement smooth but broad (slope hard to read); velocity half as wide, tilt visible; acceleration sharpest at the event, noisy elsewhere	read slopes on velocity ; use acceleration as a timing cross-check; displacement for robustness only
band-pass	5 Hz keeps bulk motion; 15–90/100 Hz narrowest clean fronts; 8–45 smears; 30–150 rings	15–100 Hz (literature band)
directional filter	imposes a tilt, including on the no-wave window; direction follows the setting	off
spatial smoothing	none striped; 0.6–1.0 mm Gaussian clean; 2×4 mm fabricates bands in noise	Gaussian 0.6–1.0 mm
temporal smoothing	no visible effect on band-passed displacement	mean 3–5 on velocity / acceleration
M-line averaging	little effect up to 9 lines; 15 lines smooths noise into bands	5–9 lines within the wall
IQ low-pass / SVD	no visible change once the high-pass is in place	not needed
polynomial detrend	edge bands, more low-frequency content	not a substitute for the band-pass
CFWI	fronts in 3–5 of 6 windows at lower contrast; direction at chance	not better here

Caveats.

- Seven windows, labelled by one observer; the confidence scores were not blind.
- The lines are short (19–22 mm) compared with the shear wavelength (150 mm or more at 20 Hz). Much of the “front” is therefore the whole segment moving nearly together, and a method can look good by making bands vertical.
- Per-panel colour scaling hides amplitude differences.
- These observations favour a velocity view in the ~ 15 –100 Hz band (close to view C and to Keijzer’s recipe) over displacement for reading slopes. That differs from the earlier “displacement is cleanest” conclusion, which was reached on one acquisition. A second observer’s reading (`docs/TODO.md`, item 4) should settle it.

Reproduce.

- `python study/analysis/passive_methods_atlas.py` reads the seven windows from `Z:/raw_data` (~ 20 min) and writes `report/passive_methods/figures/`.
- `--figures` re-renders from the cache.
- Compile with `tectonic passive_methods.tex`.